

From Bühlmann to Bayes: Hierarchical Partial Pooling and Credibility Theory in Auto Insurance Claim Frequencies

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Nicholas Bartolucci

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1 Introduction

Hierarchical Bayesian models occupy an important position in applied statistics: they are simultaneously a modeling choice and a philosophical stance on how information interacts between related groups. The core idea is simple. When estimating group-specific parameters, such as regional claim rates, school effects, and patient-level responses, the analyst faces a fundamental tension. Treating each group as entirely independent ignores the fact that groups often share a common data-generating process, discarding useful information and producing unstable estimates especially for sparse groups. Treating all groups as identical imposes a structure that is almost certainly wrong, masking heterogeneity behind a single estimate. Hierarchical models resolve this through partial pooling: each group’s estimate is shrunk toward a collective mean, with the level of shrinkage determined by the amount of data the group contributes relative to the estimated

variability across groups. Data-rich groups speak largely for themselves; data-sparse groups borrow strength from the collective.

This partial pooling mechanism has a precise classical analogue, notably, one related to my person interest in actuarial science, which inspired my choice for this project. Bühlmann credibility theory, developed in actuarial literature in the 1960s, derives the optimal estimator of a group’s true risk rate under squared error loss as a weighted combination of the group’s own experience and the collective mean, with the weight (credibility factor Z_r) determined by the ratio of within-group and between-group variance. Under certain conditions this estimator is analytically equivalent to the posterior mean of a Bayesian hierarchical model, connecting a result derived for insurance pricing to the broader topic of modern Bayesian inference used in class.

The practical difficulty with hierarchical models is that partial pooling is governed by a variance component, σ_u , which controls how much groups are allowed to deviate from the collective mean. This parameter is informed by the group-level data alone: regardless of how many individual observations are available, σ_u is identified only by the number of groups. This bottleneck has a direct consequence for Bayesian inference - the prior placed on σ_u remains influential for longer than priors on individual-level parameters, and may dominate inference entirely when a researcher lacks a sufficient amount of group-level data. Determining when prior choice on σ_u matters, and when the likelihood renders it irrelevant, is one of the central questions this project addresses.

2 Data Description

This analysis uses the `freMTPL2freq` dataset to investigate whether hierarchical Bayesian modeling improves estimation of regional auto insurance claim frequencies relative to a flat pooled model. Specifically, the research question is: does partial pooling across regions reduce posterior uncertainty in claim rate estimates, and under what data conditions does the choice of prior on regional variance matter? To answer this, three models are fit: a flat Poisson GLM, and two hierarchical Poisson GLMs differing only in their prior on σ_u . We look at these models across four sample sizes ranging from $N = 500$ to $N = 676,789$, with results connected to classical Bühlmann credibility theory.

The dataset `freMTPL2freq` is sourced from the `CASdatasets` R package and made available via Kaggle. It contains 678,013 French motor third-party liability insurance policies observed over one year. Each row represents a single policy with the following variables:

- **ClaimNb**: number of claims filed during the exposure period (response variable)
- **Exposure**: policy duration in years, used as an offset
- **DrivAge**: driver age in years
- **BonusMalus**: French no-claims discount score (50–350); values below 100 indicate a claims-free bonus, above 100 a malus
- **VehPower**: vehicle power (ordered categorical)
- **VehAge**: vehicle age in years
- **VehGas**: fuel type (Diesel or Regular), encoded as binary
- **Density**: population density of the driver’s municipality, log-transformed
- **Region**: one of 21 French administrative regions, used as the grouping variable for hierarchical modeling

Data cleaning: 1,224 policies with exposure exceeding one year were removed as they were likely data anomalies, leaving $N = 676,789$ policies for analysis. Continuous predictors were standardised to have mean zero and unit variance prior to modeling.

Table 1: Claim rates by French region

Region	Policies	Claims	Exposure	Claim Rate
Corse	4516	247	1765.935	0.140
Champagne-Ardenne	3026	160	1205.832	0.133
Ile-de-France	69791	3978	30208.642	0.132
Picardie	7994	439	3574.410	0.123
Rhone-Alpes	84752	5032	45347.417	0.111
Alsace	2200	133	1209.116	0.110
Provence-Alpes-Cotes-D’Azur	79315	3907	35790.321	0.109
Limousin	4567	259	2395.995	0.108
Languedoc-Roussillon	35805	1547	14736.104	0.105
Bourgogne	10492	511	5025.359	0.102
Franche-Comte	1326	57	564.013	0.101
Midi-Pyrenees	17141	688	7020.109	0.098
Bretagne	42122	2702	27753.464	0.097
Haute-Normandie	8784	303	3177.682	0.095
Basse-Normandie	10893	633	6657.901	0.095
Aquitaine	31329	1348	14322.167	0.094
Pays-de-la-Loire	38751	2010	21934.031	0.092
Nord-Pas-de-Calais	40275	1787	19611.752	0.091
Centre	160601	9204	102712.903	0.090
Poitou-Charentes	19046	968	11163.409	0.087
Auvergne	5287	189	2322.884	0.081

Global claim rate: 0.1007

Table 1 reveals substantial heterogeneity in claim rates across the 21 French regions, ranging from 0.081 in Auvergne to 0.140 in Corse, creating a 72% spread around the global mean of 0.1007. Critically, this variation is confounded with portfolio size: Corse and Champagne-Ardenne, the two highest-rate regions, have only 4,516 and 3,026 policies respectively, while Centre, the largest region, contributes 160,601 policies at a below-average rate of 0.090. Raw claim rates for small regions are therefore unreliable estimates of true underlying risk, being highly sensitive to a handful of claims. This tension between data sparsity and regional heterogeneity motivates the hierarchical modeling approach developed in our analysis, where partial pooling allows small regions to borrow strength from the collective experience while large regions are largely governed by their own data.

To tackle the research question, three models are fit and compared: a flat Poisson GLM imposing complete pooling across regions, a hierarchical Poisson GLM with a weakly informative prior on the regional variance component $\sigma_u \sim \text{Half} - \text{Normal}(0, 0.5)$, and a hierarchical Poisson GLM with an informative prior $\sigma_u \sim \text{Half} - \text{Normal}(0, 0.1)$. Although a gamma prior is a natural choice for model conjugacy regarding precision parameters, we selected a half-normal prior for computational behaviour, and direct interpretability regarding variation. This is often recommended by Gelman et al. (2006) and is standard practice for hierarchical stan models. All three share identical priors on the intercept and fixed effects. To isolate the role of sample size in prior sensitivity, each model is refit on four subsets of the data: $N = 500$, $N = 2,000$, $N = 10,000$, and the full $N = 676,789$. Posterior uncertainty in regional claim rate estimates is the primary quantity of interest, with shrinkage behaviour and prior sensitivity assessed as a function of N . Results are further connected to classical Bühlmann credibility theory to provide an actuarial interpretation of the partial pooling mechanism.

3 Prior Choice

We adopt weakly informative priors throughout:

$$\beta_j \sim N(0, 1), \quad j = 1, \dots, 6$$

Using the Poisson model, this places 95% prior mass between $e^{-2} \approx 0.14$ and $e^2 \approx 7.4$, wide enough to be non-informative in practice.

For the intercept we incorporate actuarial industry knowledge. The global claim rate is approximately 0.10 claims per exposure year, implying a log baseline of $\log(0.10) \approx -2.3$. We therefore set:

$$\alpha \sim \mathcal{N}(-2, 0.5)$$

placing 95% prior mass on baseline rates between $e^{-3} \approx 0.05$ and $e^{-1} \approx 0.37$, which is consistent with typical motor insurance claim frequencies.

The flat Poisson GLM uses identical priors on the intercept and fixed effects, with $\alpha \sim \mathcal{N}(-2, 0.5)$ and $\beta_j \sim N(0, 1)$, but contains no regional structure and therefore no variance component. This acts as a complete pooling baseline for which the hierarchical models will be compared.

The parameter σ_u controls heterogeneity across the 21 French regions. Critically, σ_u is informed by only 21 group-level estimates regardless of total N , hence this is where prior choice genuinely matters. We compare two priors:

Prior 1 — Weakly Informative:

$$\sigma_u \sim \text{Half-Normal}(0, 0.5)$$

This allows regions to differ substantially if supported by the data, with $\sigma_u = 0.5$ implying regional rate ratios of up to $e^{0.5} \approx 1.65$ vs the global mean.

Prior 2 — Informative:

$$\sigma_u \sim \text{Half-Normal}(0, 0.1)$$

With this we express the belief that regions are largely similar, placing 95% of prior mass below $\sigma_u \approx 0.196$. On the credibility scale this corresponds to a strong prior belief that regional observations should be heavily discounted in favour of collective experience, equivalent to assuming low Bühlmann credibility factors Z_r .

Whether the prior choice materially affects inference depends on how much data is available to estimate σ_u . Our later analysis investigates this directly by refitting all three models across sample sizes ranging from $N = 500$ to $N = 676,789$, revealing precisely when the prior on σ_u dominates and when the likelihood renders it irrelevant.

4 Analysis

Three models are compared across four sample sizes to isolate the conditions under which hierarchical modeling and prior choice materially affect inference. The complete pooling model ignores regional structure entirely, estimating a single baseline adjusted for covariates. The two hierarchical models introduce region-level random effects that shrink regional estimates toward the global mean, differing only in the variance of the prior placed on σ_u - the parameter governing how much regions are permitted to deviate from that mean. For each sample size, three outputs are presented: a shrinkage plot showing how each model's posterior regional rates relate to the raw rates, a summary table quantifying posterior uncertainty, and a density plot comparing the two hierarchical posteriors for σ_u .

The hierarchical model introduces region-level random effects u_r , allowing claim rates to vary across the 21 French regions while borrowing information from partial pooling. The model is specified as:

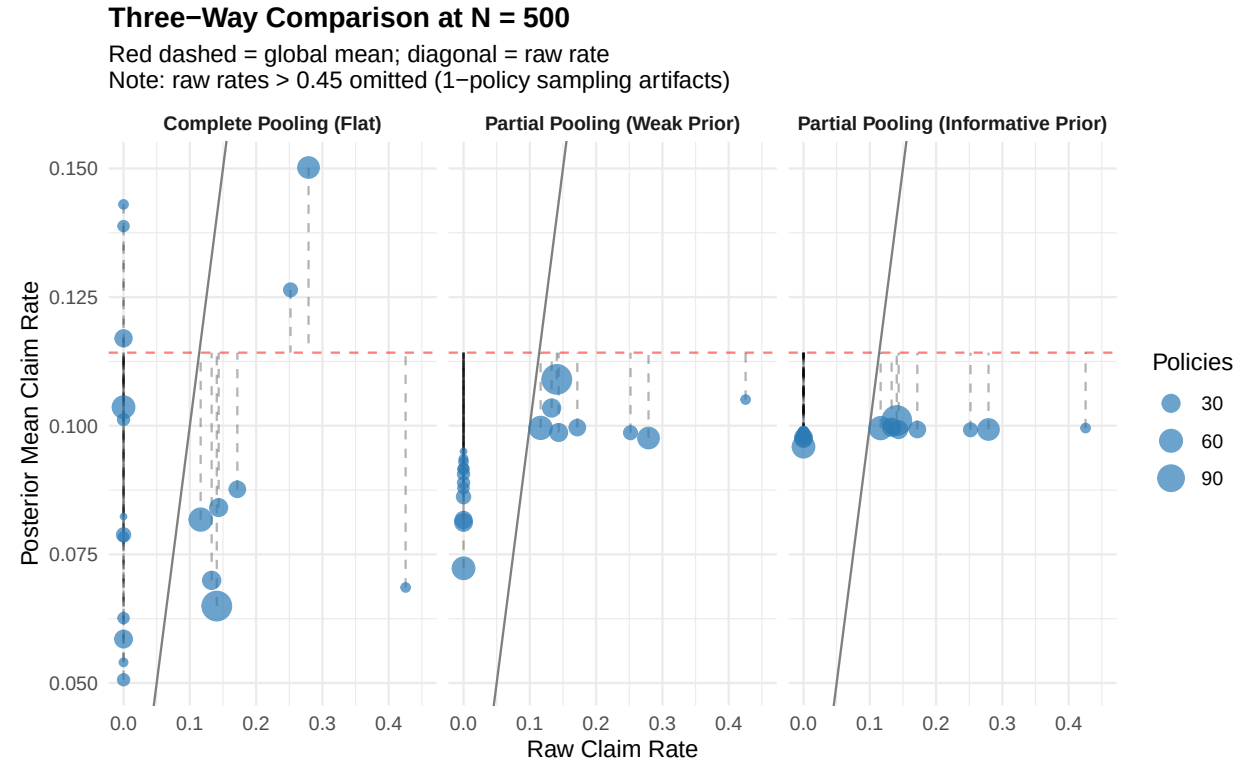
$$\begin{aligned} Y_i &\sim \text{Poisson}(\lambda_i \cdot e_i) \\ \log(\lambda_i) &= \alpha + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{r[i]} \\ u_r &\sim \mathcal{N}(0, \sigma_u^2), \quad r = 1, \dots, 21 \end{aligned}$$

A non-centered parameterisation is employed for numerical stability:

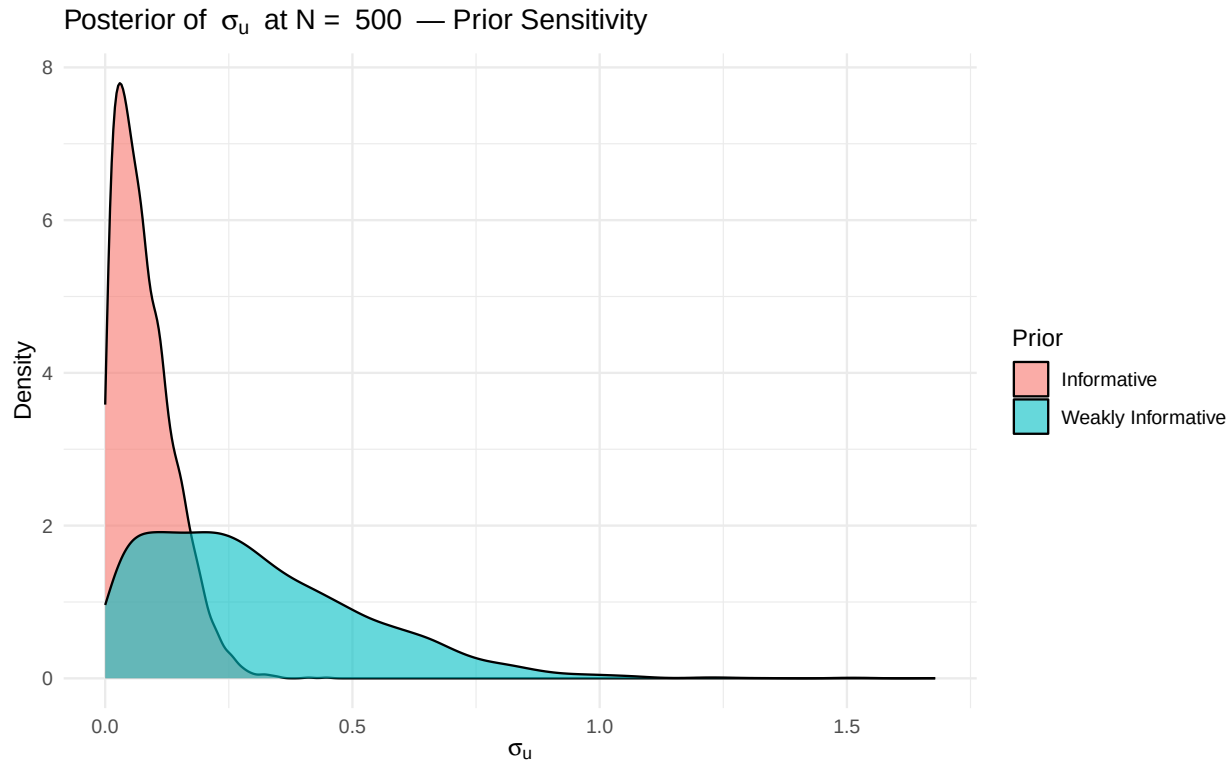
$$u_r = \sigma_u \cdot \tilde{u}_r, \quad \tilde{u}_r \sim \mathcal{N}(0, 1)$$

We note that the standard deviations reported by the models are fundamentally different objects. In the complete pooling model, every region shares a single global intercept, so the posterior SD reflects uncertainty about that one scalar parameter, being the same value replicated across all 21 regions. In the hierarchical models, the posterior SD is region-specific: it reflects uncertainty about each region's deviation from the global mean, a parameter that must be estimated from that region's data alone. A smaller posterior SD in the flat model therefore does not indicate greater precision about regional risk; it indicates that the flat model has collapsed all regional heterogeneity into a single number and is expressing confidence about that number, while remaining silent about genuine between-region variation. Similarly, the range of posterior means measures something different in each case: for the flat model it reflects covariate-driven differences in regional profiles, while for the hierarchical models it reflects the regional effects after partial pooling toward the global mean. Both metrics are informative within each model, but cross-model comparisons should be read as descriptions of what each model is doing rather than a strict precision ranking.

4.1 Small sample, N=500



```
##
##
## Table: Posterior Uncertainty at N = 500 : Three-Way Comparison
##
## |Model| Mean Posterior SD| Range of Posterior Means|
## |-----|-----|-----|
## |Complete Pooling (Flat)| 0.2346| 0.0996|
## |Partial Pooling (Weak Prior)| 0.3419| 0.0367|
## |Partial Pooling (Informative Prior)| 0.0991| 0.0051|
```



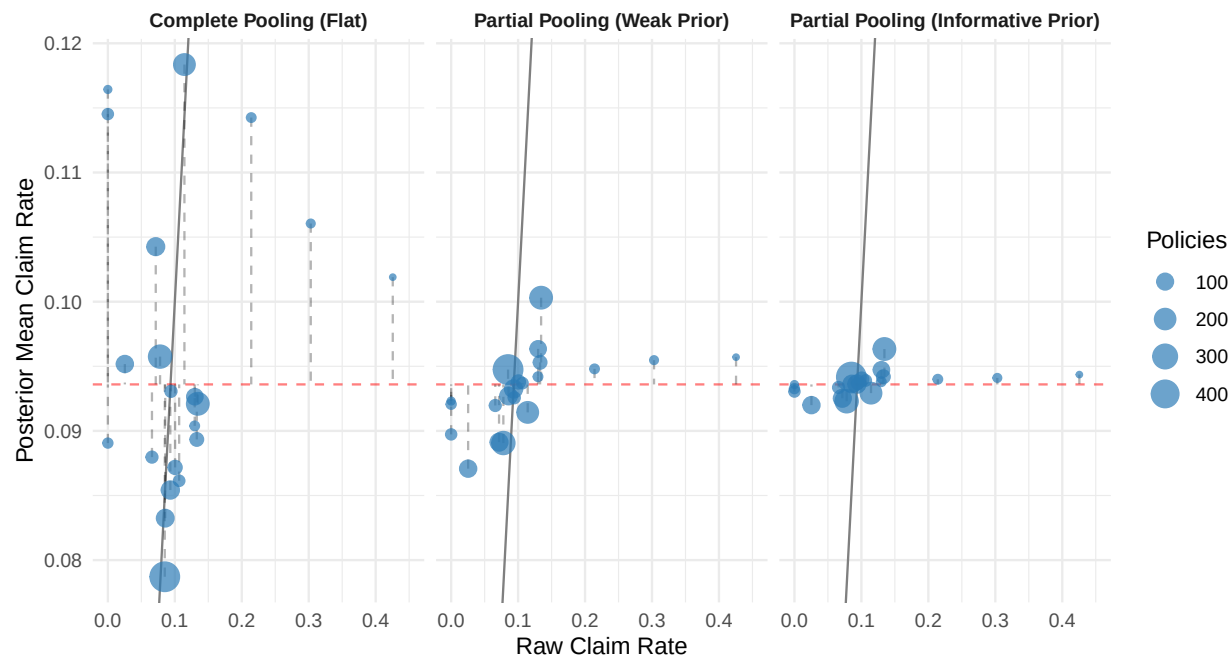
At $N = 500$, with proportional stratified sampling giving Centre roughly 119 policies and Franche-Comté only 1, raw claim rates are highly unstable, ranging from 0.10 to over 0.34 across regions. The complete pooling model partially recovers regional variation through covariate adjustment, producing a range of posterior means of 0.100, but with no shrinkage mechanism its estimates track the covariate-driven signal regardless of how little data underlies each region. The weak hierarchical prior reduces this range substantially to 0.037, reflecting the stabilising effect of partial pooling as regions borrow strength from collective experience. The informative prior goes considerably further, collapsing the range to just 0.005 — essentially imposing complete pooling in practice. At this sample size the likelihood over σ_u is too weak to overcome the prior belief that regions are similar, so the informative model discards nearly all regional signal. The σ_u density plot makes this explicit: the two hierarchical posteriors are visibly separated, with the informative prior concentrating near zero while the weak prior allows for a long right tail. Whether the shrinkage of the informative prior is desirable depends entirely on whether the actuarial belief that French regions are largely homogeneous in risk is actually correct. Crucially, this question is hardest to answer precisely when it matters most: at small N , where the data cannot adjudicate between the two priors, the analyst’s belief about regional heterogeneity is the dominant driver of inference.

4.2 Moderate sample, N=2000

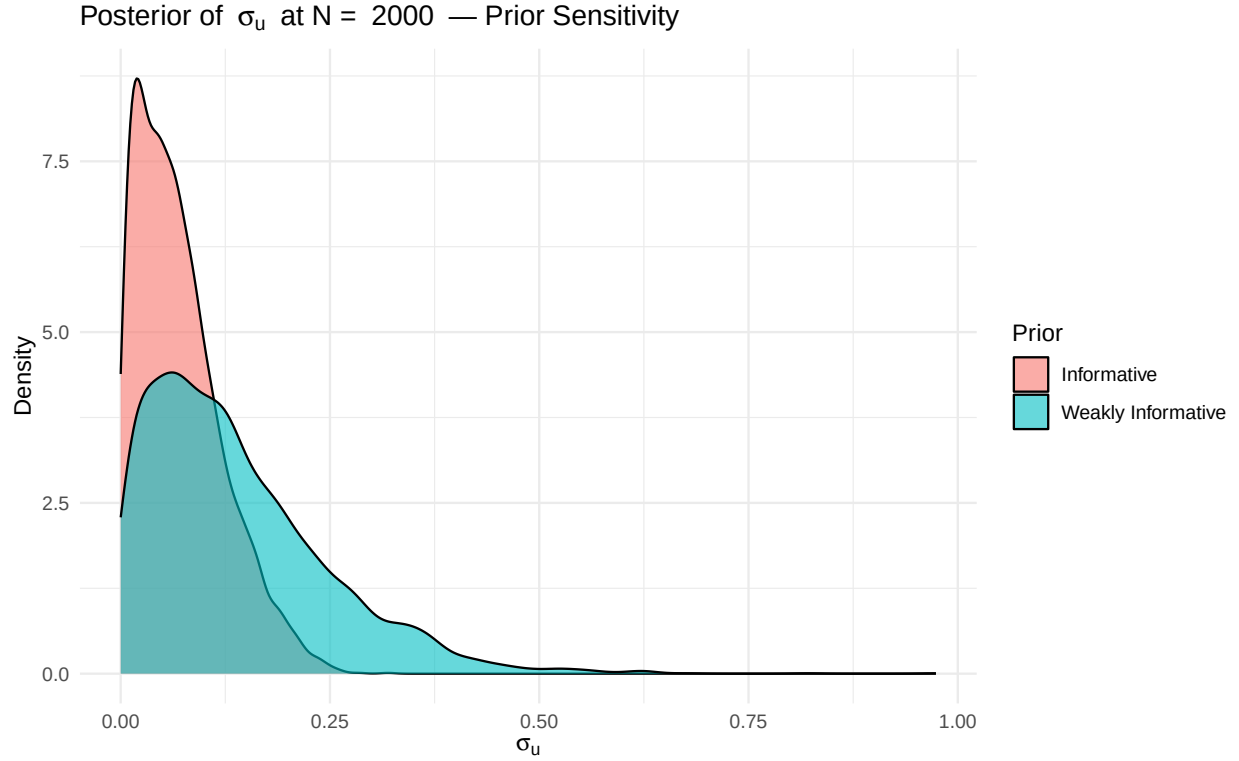
Three-Way Comparison at N = 2000

Red dashed = global mean; diagonal = raw rate

Note: raw rates > 0.45 omitted (1-policy sampling artifacts)



```
##
##
## Table: Posterior Uncertainty at N = 2000 : Three-Way Comparison
##
## |Model| Mean Posterior SD| Range of Posterior Means|
## |-----|-----|-----|
## |Complete Pooling (Flat)| 0.1361| 0.0396|
## |Partial Pooling (Weak Prior)| 0.1582| 0.0132|
## |Partial Pooling (Informative Prior)| 0.0838| 0.0043|
```



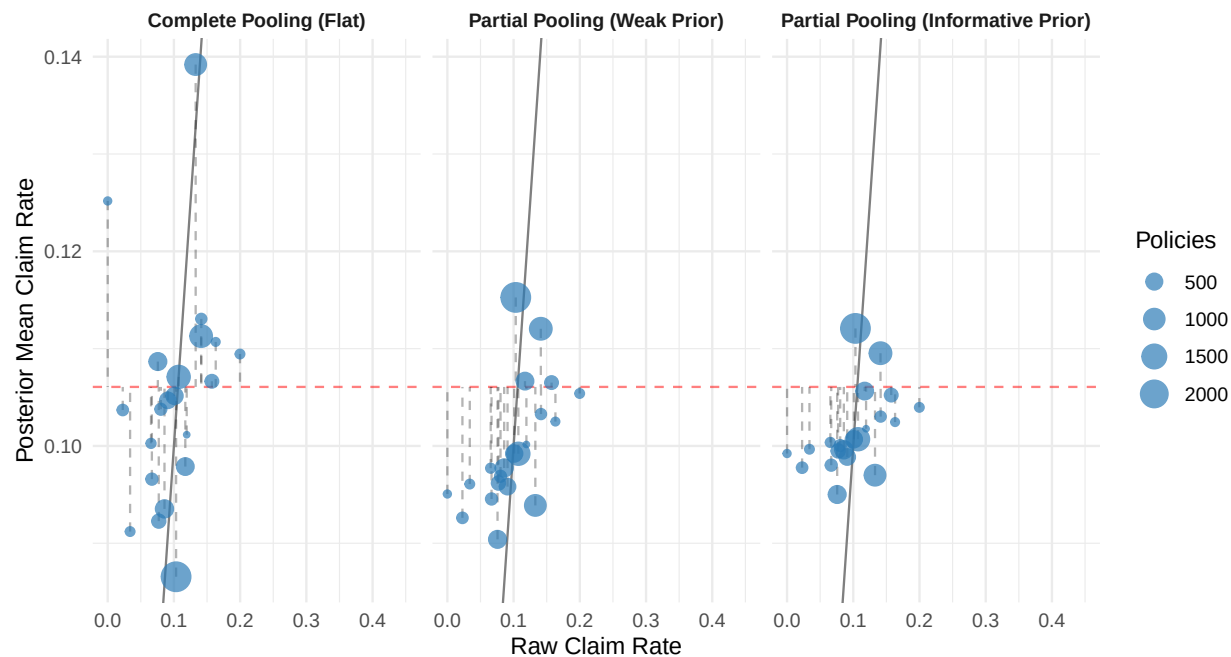
At $N = 2,000$, the raw rates stabilise somewhat but remain noisy. The key development relative to $N = 500$ is that the two hierarchical models begin to converge: the gap between the informative prior range (0.004) and the weak prior range (0.013) has narrowed from 0.032 at $N = 500$ to 0.009 here, as the likelihood accumulates enough information to begin overriding the tight prior on σ_u . The σ_u density plot shows the posteriors still clearly separated, but the informative prior's peak has shifted slightly rightward as the data pushes back against the prior belief that regions are homogeneous. The hierarchical model now produces a tighter distribution of regional effects around the global mean, with covariate-driven heterogeneity in the flat model no longer the dominant source of cross-region spread.

4.3 Large sample, N=10000

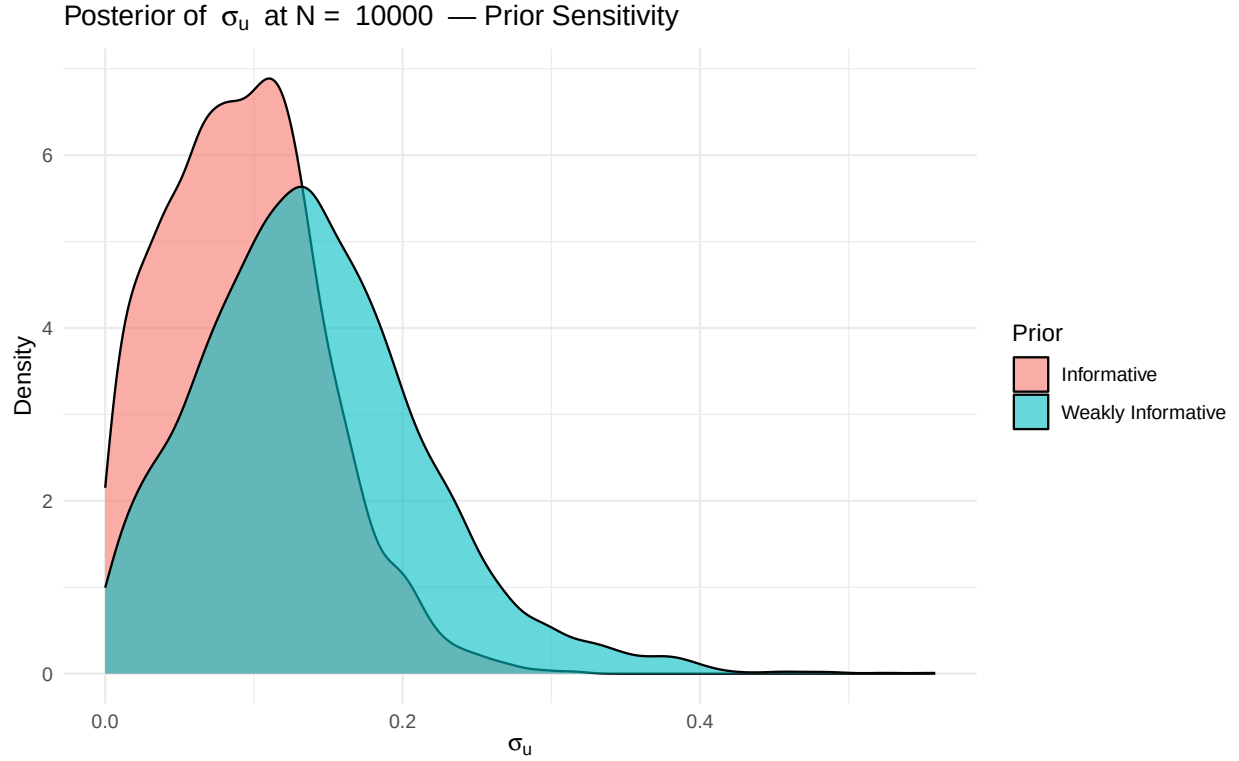
Three-Way Comparison at N = 10000

Red dashed = global mean; diagonal = raw rate

Note: raw rates > 0.45 omitted (1-policy sampling artifacts)



```
##
##
## Table: Posterior Uncertainty at N = 10000 : Three-Way Comparison
##
## |Model| Mean Posterior SD| Range of Posterior Means|
## |-----|-----|-----|
## |Complete Pooling (Flat)| 0.0592| 0.0526|
## |Partial Pooling (Weak Prior)| 0.1333| 0.0249|
## |Partial Pooling (Informative Prior)| 0.0968| 0.0171|
```

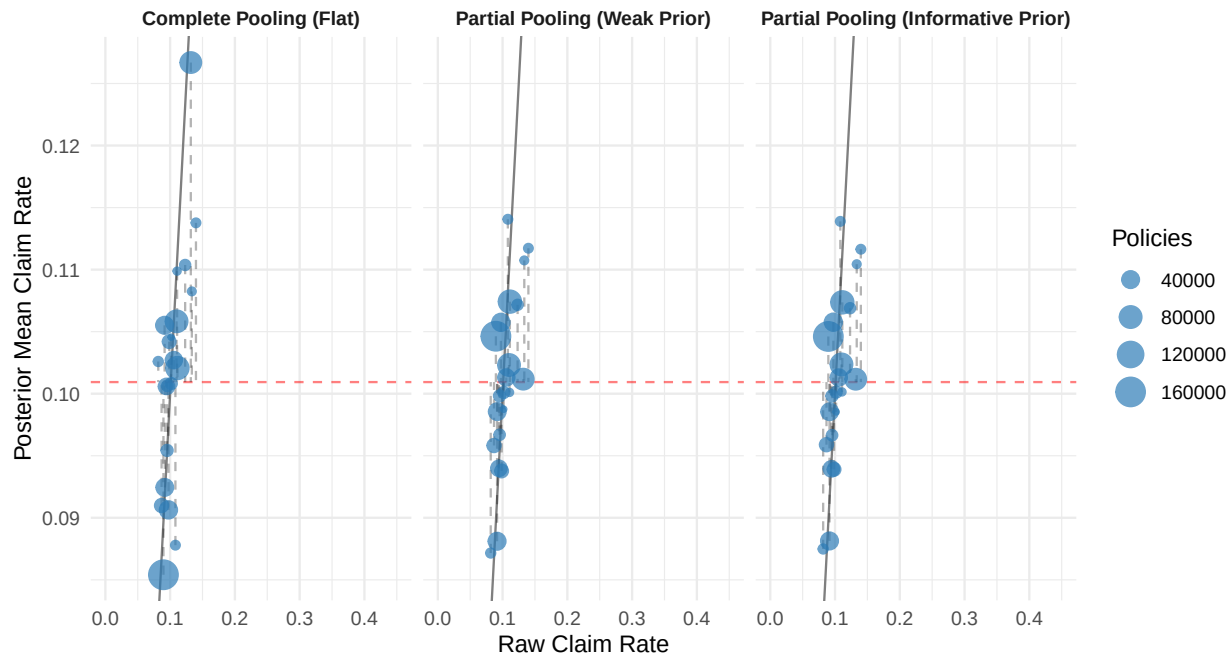


At $N = 10,000$ proportional allocation gives Centre over 2,300 policies while the smallest regions still receive only a handful, and the likelihood is strong enough to meaningfully inform σ_u under both priors. The gap between the two hierarchical models continues to close: the informative prior range (0.017) is now approaching the weak prior range (0.025), compared to a gap of 0.009 at $N = 2,000$ and 0.032 at $N = 500$. The σ_u density plot shows the two posteriors overlapping significantly, with both concentrated in the (0.1, 0.3) range — the data is actively overriding the prior's preference for near-zero regional variance. The weak hierarchical model's shrinkage plot now tracks the diagonal more closely than at smaller N , reflecting that the larger regions have garnered enough data to resist excessive pooling, while the smallest regions remain appropriately shrunk toward the global mean. The size variation introduced by proportional sampling is now visible in the shrinkage plots: larger regions sit closer to the diagonal while smaller regions are pulled more strongly toward the global mean, a pattern that was invisible under the original equal-stratification design.

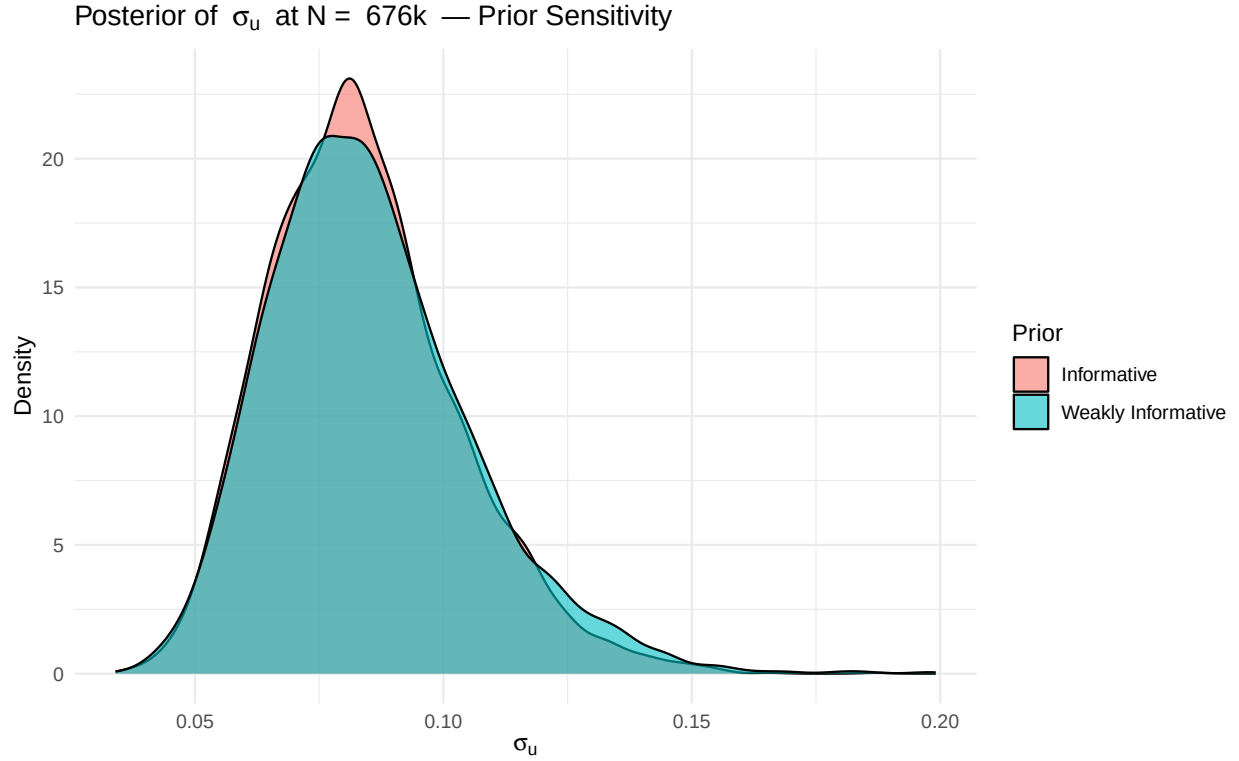
4.4 Full sample, N=676k

Three-Way Comparison at N = 676k

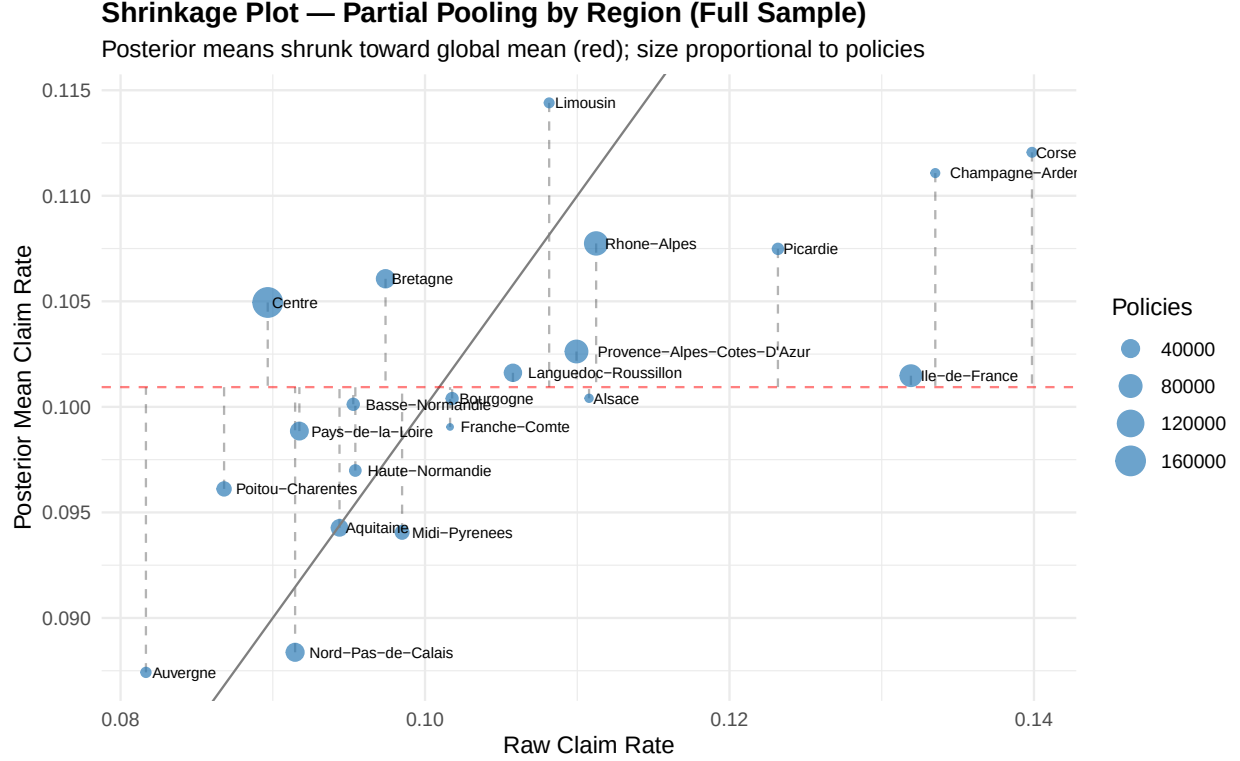
Red dashed = global mean; diagonal = raw rate
Note: raw rates > 0.45 omitted (1-policy sampling artifacts)



```
##
##
## Table: Posterior Uncertainty at N = 676k : Three-Way Comparison
##
## |Model| Mean Posterior SD| Range of Posterior Means|
## |:-----:|:-----:|:-----:|
## |Complete Pooling (Flat)| 0.0073| 0.0413|
## |Partial Pooling (Weak Prior)| 0.0416| 0.0269|
## |Partial Pooling (Informative Prior)| 0.0413| 0.0264|
```



With $N = 676,789$ observations the likelihood completely dominates both priors. The two hierarchical models are now virtually indistinguishable: posterior ranges of 0.027 and 0.026, and mean posterior SDs of 0.042 and 0.041 respectively. The σ_u density plot shows nearly complete overlap, with posterior means of 0.085 (weak) and 0.084 (informative), a negligible difference. Prior choice on σ_u is therefore inconsequential at full sample size; any reasonable prior would yield the same result. The complete pooling model retains a range of 0.041 driven purely by covariate heterogeneity across regions, but remains blind to regional structure. The labeled shrinkage plot below shows the full-sample hierarchical estimates by region, illustrating how partial pooling allocates trust between regional experience and collective knowledge as a function of portfolio size.



Taken together, the four sample sizes tell a coherent story about the interplay between data, prior, and pooling. At $N = 500$ the informative prior dominates, effectively imposing complete pooling regardless of the model specification — the likelihood is too weak to resist it. By $N = 2,000$ the data begins to push back, and by $N = 10,000$ the two hierarchical models are converging as the likelihood progressively overwhelms the prior on σ_u . At full sample size the distinction between the two priors is negligible. The practical implication is clear: prior choice on σ_u matters most precisely when data is scarcest, the small portfolio setting where actuarial judgment is most needed and least verifiable. In such settings the choice between a weakly informative and an informative prior is not a technical detail but an important decision about how much regional heterogeneity the analyst is willing to believe exists before seeing the data.

5 Convergence Diagnostics

Convergence diagnostics were assessed using three standard criteria: the $\hat{R}$ values, effective sample size, and sampler divergences. All twelve tests (three models across four sample sizes) satisfy the criteria, giving an analysis that rests on well-mixed, well-identified posterior distributions. We also note that, in our simulations default Stan HMC settings (adapt_delta = 0.8, max_tredepth = 10, 1000 warmup + 1000 sampling iterations per chain, 4 chains) produced no divergences or depth warnings at any sample size; so no tuning adjustments were required.

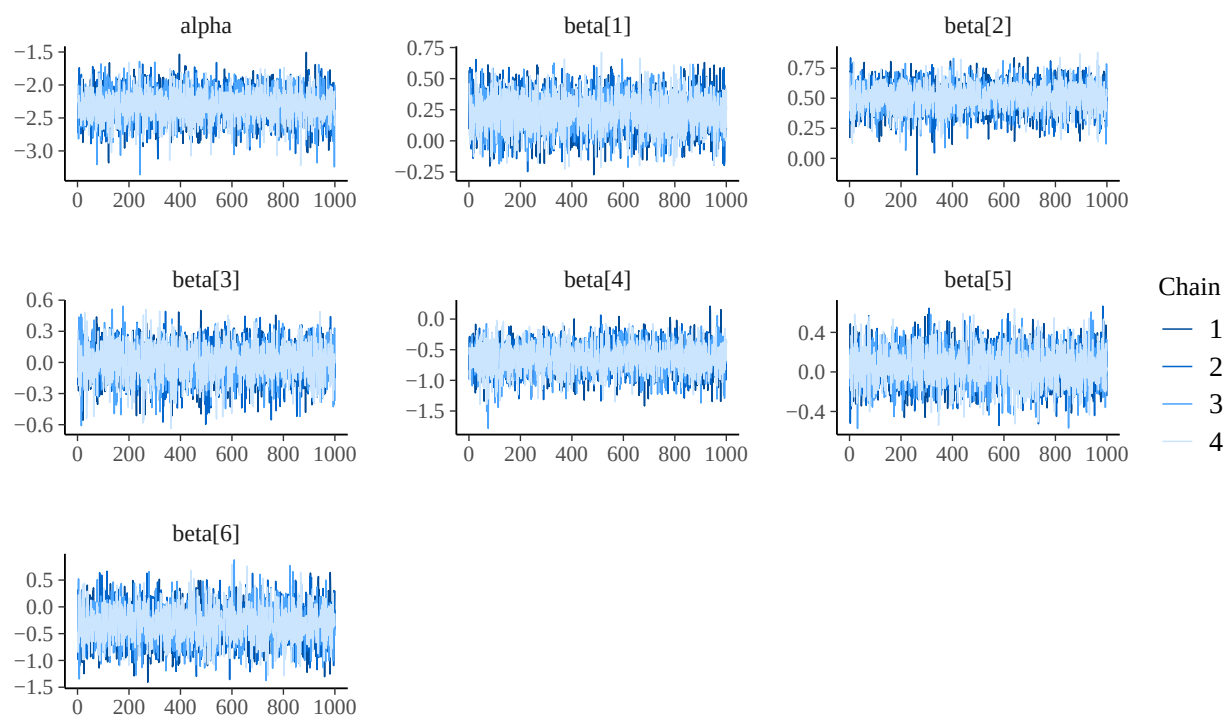
Table 2: Convergence diagnostics across all models and sample sizes

N	Model	Max Rhat	Min ESS		
			Bulk	Min ESS Tail	Divergences
500	Complete Pooling (Flat)	1.003	3882	2786	0
500	Partial Pooling (Weak Prior)	1.002	1781	2134	0

N	Model	Max	Min ESS	Min ESS Tail	Divergences
		Rhat	Bulk		
500	Partial Pooling (Informative Prior)	1.003	3604	2375	0
2000	Complete Pooling (Flat)	1.003	3772	2853	0
2000	Partial Pooling (Weak Prior)	1.003	1975	1825	0
2000	Partial Pooling (Informative Prior)	1.003	3470	2075	0
10000	Complete Pooling (Flat)	1.002	3797	2726	0
10000	Partial Pooling (Weak Prior)	1.003	1200	1225	0
10000	Partial Pooling (Informative Prior)	1.005	1571	1452	0
676k	Complete Pooling (Flat)	1.001	3452	2523	0
676k	Partial Pooling (Weak Prior)	1.003	706	1194	0
676k	Partial Pooling (Informative Prior)	1.005	730	1188	0

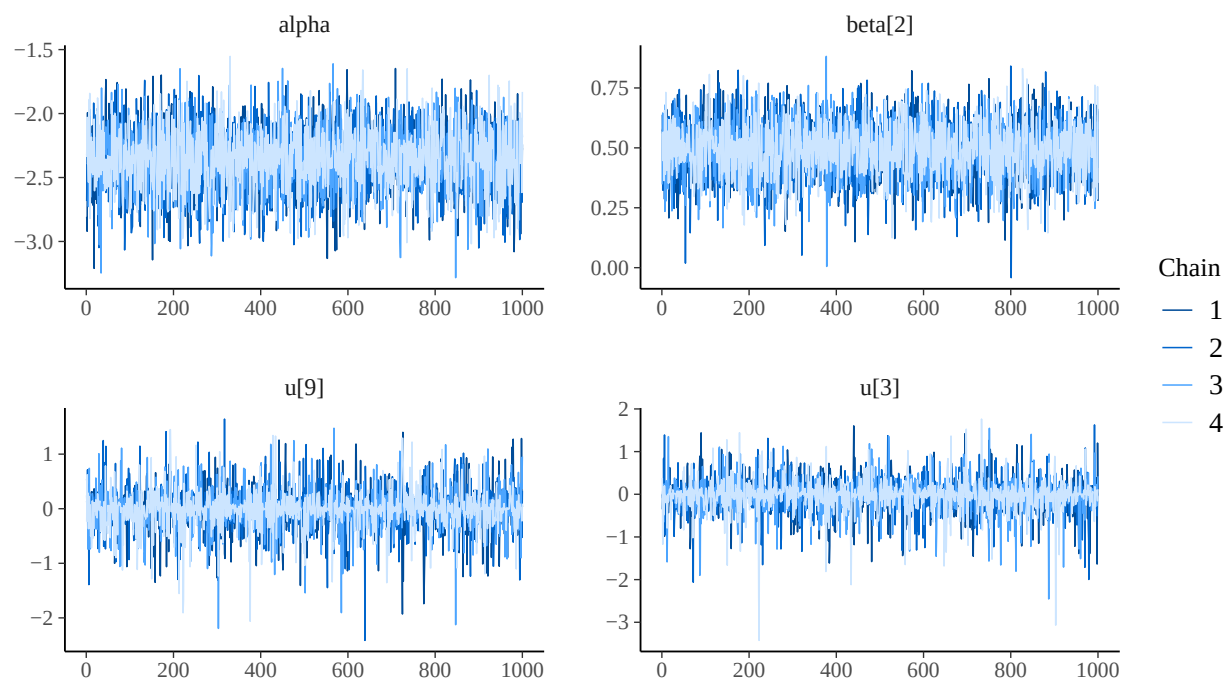
Table 2 confirms clean convergence throughout. Every $\hat{R}$ falls at or below 1.005, no divergences are recorded at any sample size or model specification, and all ESS values, bulk and tail, comfortably exceed 400. Two structural patterns are visible but neither indicates a sampling problem. First, hierarchical models consistently produce lower ESS bulk than the flat model at the same N : this reflects the additional posterior geometry introduced by 21 region-level parameters and their shared variance component σ_u , which creates mild funnel-shaped dependencies that a flat model never encounters. Notably, ESS tail for the hierarchical models often exceeds ESS bulk, especially at $N = 500$ where the weak prior gives bulk 1,781 but tail 2,134, which is the opposite of the pattern one would expect if the funnel geometry were causing tail mixing problems. This confirms that the non-centered parameterisation $u_r = \sigma_u \tilde{u}_r$ is doing its job: the sampler explores the tails of σ_u efficiently even at small N , and no pathological tail behaviour is present. Second, ESS tends to decrease as N grows, particularly for the hierarchical models at $N = 676k$. This is a geometry effect, not a sampler failure since at very large N the posterior concentrates sharply and the chains must take finer steps, reducing the number of effectively independent draws per iteration. The values of 706 and 730 at full sample size remain above the threshold for reliable posterior summarisation and their tail ESS values of 1,194 and 1,188 confirm that inference on σ_u is equally reliable.

Trace Plots — Complete Pooling (Flat), N = 500



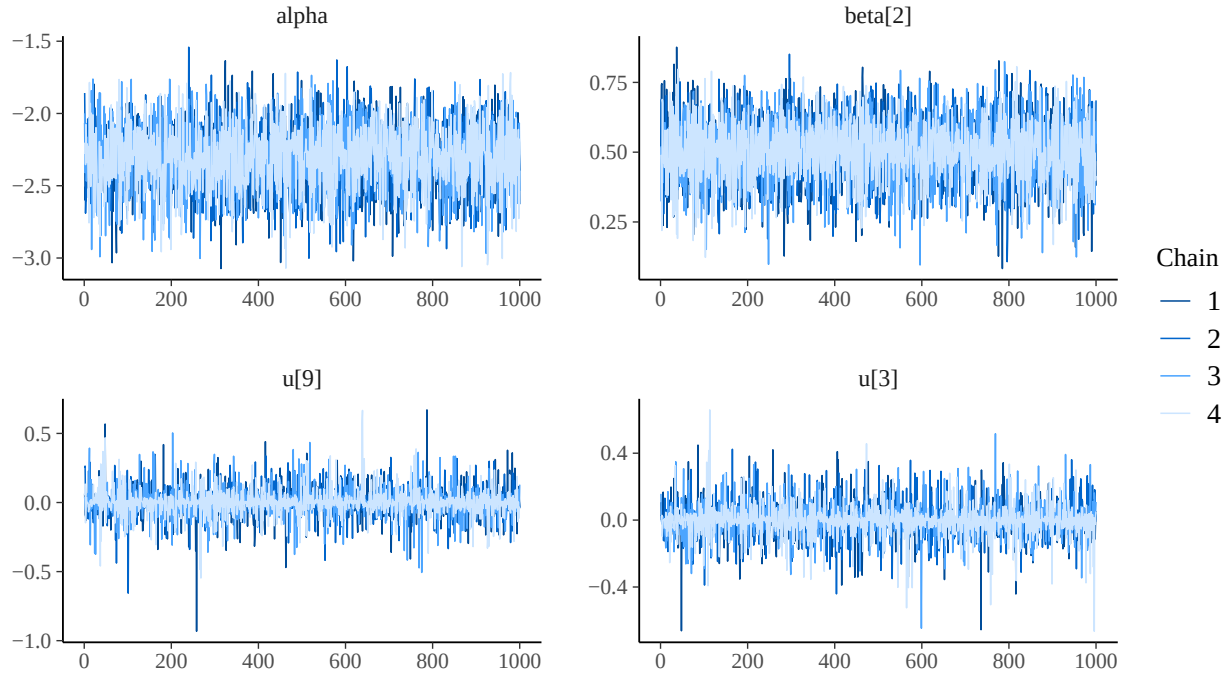
Trace Plots — Weak Prior, N = 500

$u[9]$ = Corse, $u[3]$ = Auvergne. σ_u shown separately.

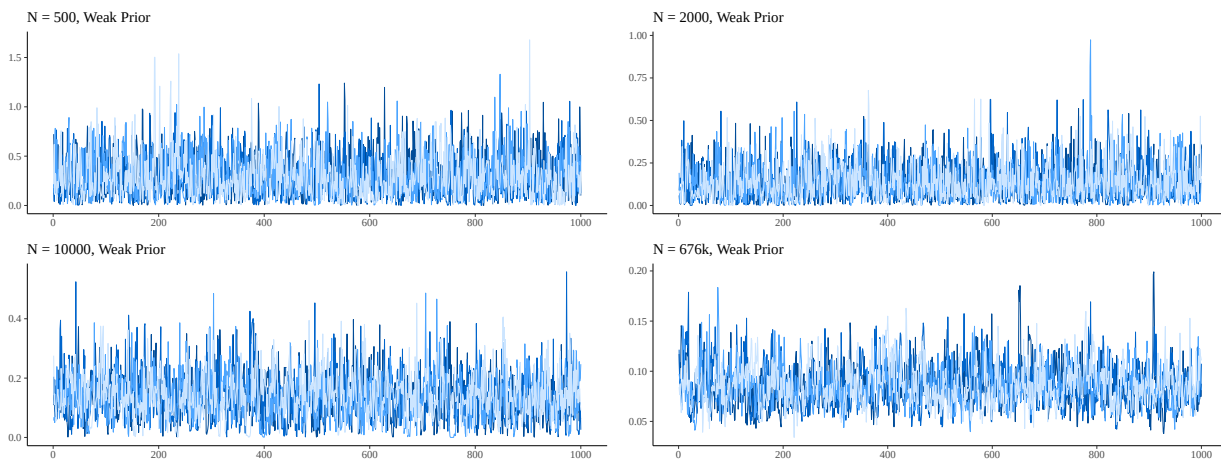


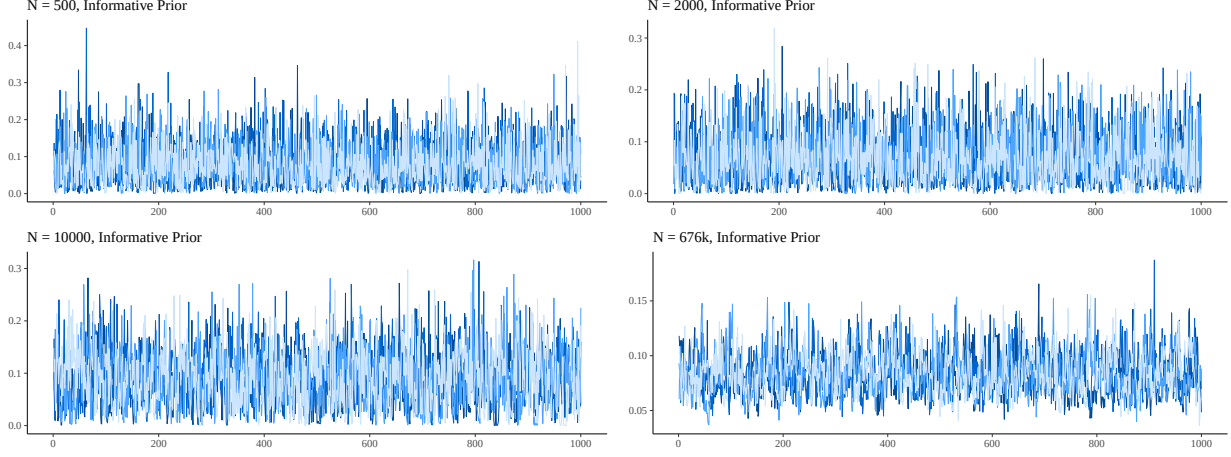
Trace Plots — Informative Prior, $N = 500$

$u[9] = \text{Corse}$, $u[3] = \text{Auvergne}$. σ_u shown separately.



The trace plots at $N = 500$ reflect the theoretical properties of each model. The flat model's parameters mix cleanly by construction: with no group structure to estimate, the posterior geometry is simple and the four chains overlap immediately. The hierarchical parameters (global intercept, fixed effects, regional deviations) also mix well, with chains overlapping and traversing the full posterior support without sticking. The regional effects for Corse (u_9) and Auvergne (u_3) are shown as illustrative examples; σ_u traces are displayed separately in the panels below, as its mixing behaviour across sample sizes warrants dedicated attention.





The σ_u traces tell a coherent story across all eight panels. Because σ_u is identified by only 21 group-level estimates regardless of total N , the posterior retains a mild funnel geometry throughout—near-zero values of σ_u compress the regional deviations toward a degenerate surface independent of how much individual-level data accumulates. The non-centered parameterisation $u_r = \sigma_u \tilde{u}_r$ directly addresses this geometry, which is why no divergences are recorded even at $N = 500$. As N grows, the likelihood becomes more informative about the regional spread, the posterior for σ_u concentrates away from zero, the funnel neck is cut off by the data, and the near-zero excursions disappear. By $N = 676k$ both sets of chains are tight, stationary, and indistinguishable from one another; consistent with the $\hat{R}$ and ESS values in Table 2 and with the convergence observed in our analysis.

Table 3: Posterior Predictive Summary, $N = 500$

Model	Observed Mean	Predicted Mean	Observed Variance	Predicted Variance	Observed P(Y=0)	Predicted P(Y=0)
Complete Pooling (Flat)	0.0621	0.0595	0.0744	0.0707	0.9459	0.9463
Partial Pooling (Weak Prior)	0.0621	0.0560	0.0744	0.0647	0.9459	0.9492
Partial Pooling (Informative Prior)	0.0621	0.0588	0.0744	0.0721	0.9459	0.9473

Table 4: Posterior Predictive Summary, $N = 2000$

Model	Observed Mean	Predicted Mean	Observed Variance	Predicted Variance	Observed P(Y=0)	Predicted P(Y=0)
Complete Pooling (Flat)	0.0495	0.0485	0.0571	0.0494	0.9545	0.9531
Partial Pooling (Weak Prior)	0.0495	0.0476	0.0571	0.0489	0.9545	0.9541
Partial Pooling (Informative Prior)	0.0495	0.0489	0.0571	0.0496	0.9545	0.9526

Table 5: Posterior Predictive Summary, N = 10000

Model	Observed Mean	Predicted Mean	Observed Variance	Predicted Variance	Observed P(Y=0)	Predicted P(Y=0)
Complete Pooling (Flat)	0.0561	0.0558	0.0611	0.0583	0.9476	0.9468
Partial Pooling (Weak Prior)	0.0561	0.0558	0.0611	0.0583	0.9476	0.9468
Partial Pooling (Informative Prior)	0.0561	0.0558	0.0611	0.0584	0.9476	0.9468

Table 6: Posterior Predictive Summary, N = 676k

Model	Observed Mean	Predicted Mean	Observed Variance	Predicted Variance	Observed P(Y=0)	Predicted P(Y=0)
Complete Pooling (Flat)	0.0533	0.0532	0.0577	0.0552	0.9497	0.9491
Partial Pooling (Weak Prior)	0.0533	0.0533	0.0577	0.0553	0.9497	0.9490
Partial Pooling (Informative Prior)	0.0533	0.0532	0.0577	0.0552	0.9497	0.9491

The posterior predictive checks confirm appropriate fit across all models and sample sizes. All three models closely reproduce the observed claim mean, variance, and proportion of zero-claim policies at every N , with predicted quantities barely differing from observed counterparts. This near-perfect predictive adequacy holds regardless of prior choice on σ_u : the informative and weak priors produce nearly identical predictive distributions even at $N = 500$, where they differ most sharply in their posteriors for σ_u itself. The implication is that prior sensitivity in this analysis is confined entirely to the question of how much regional diversity the model attributes to real structural differences versus noise; it does not spread into overall predictive performance, which is governed mainly by the shared covariate structure and the global claim rate.

6 Bühlmann Credibility

A foundational result in actuarial credibility theory is the **Bühlmann credibility estimator** (Bühlmann, 1967). Given claim data from G groups, the optimal estimator of group g 's true claim rate under squared error loss is:

$$\hat{\mu}_g = Z_g \bar{Y}_g + (1 - Z_g) \bar{Y}$$

where the credibility factor is:

$$Z_g = \frac{n_g}{n_g + k}, \quad k = \frac{\sigma_{\text{within}}^2}{\sigma_{\text{between}}^2}$$

Here $Z_g \rightarrow 1$ when a group has abundant data, trusting the group's own experience, while $Z_g \rightarrow 0$ when data is sparse, shrinking toward the collective mean. This is precisely the partial pooling behaviour of the Bayesian hierarchical model. Under certain conditions the posterior mean of the hierarchical model is analytically equivalent to the Bühlmann credibility estimator, connecting a 1967 actuarial result to modern Bayesian inference.

To connect the Bühlmann framework to the fitted models, k must be estimated from the data. The within-group variance σ_{within}^2 is the average Poisson variance across regions. Since claims are Poisson, this equals the average regional claim rate. The between-group variance $\sigma_{\text{between}}^2$ is the variance of regional raw rates around the global mean, corrected for the sampling noise each regional estimate carries. This correction is necessary because raw regional rates are noisy estimates of true underlying rates, and variance of raw rates overstates true heterogeneity.

Equivalence is approximate here for two reasons. First, the hierarchical model conditions on six policy-level covariates (driver age, bonus-malus score, vehicle characteristics, population density) while classical Bühlmann operates on raw group means with no covariate adjustment. Second, analytical equivalence requires a Poisson-gamma conjugate structure, whereas the model uses a normal random effect on the log scale. What survives is the qualitative mechanism: both frameworks shrink sparse groups toward the collective mean and trust data-rich groups to speak for themselves, with the degree of shrinkage governed by the same ratio of within-group to between-group variance.

Table 7: Bühlmann vs. Hierarchical Posterior Means at $N = 500$

Region	n	Z_r	Raw Rate	Bühlmann	Hierarchical	Difference
Centre	119	0.9666	0.1407	0.1398	0.1090	-0.0308
Rhone-Alpes	62	0.9271	0.1163	0.1162	0.0996	-0.0166
Provence-Alpes-Cotes-D’Azur	58	0.9042	0.0000	0.0109	0.0723	0.0614
Ile-de-France	51	0.8883	0.2791	0.2607	0.0976	-0.1630
Bretagne	31	0.8930	0.1329	0.1309	0.1034	-0.0275
Nord-Pas-de-Calais	30	0.8375	0.1435	0.1387	0.0987	-0.0400
Pays-de-la-Loire	29	0.8699	0.0000	0.0149	0.0812	0.0664
Languedoc-Roussillon	26	0.8053	0.0000	0.0222	0.0817	0.0594
Aquitaine	23	0.8117	0.1717	0.1608	0.0996	-0.0612
Poitou-Charentes	14	0.7292	0.0000	0.0309	0.0862	0.0553
Midi-Pyrenees	13	0.5949	0.2519	0.1961	0.0986	-0.0975
Basse-Normandie	8	0.6373	0.0000	0.0414	0.0906	0.0492
Bourgogne	8	0.5711	0.0000	0.0490	0.0889	0.0400
Haute-Normandie	6	0.5881	0.0000	0.0470	0.0916	0.0445
Picardie	6	0.5618	0.0000	0.0500	0.0879	0.0378
Auvergne	4	0.4325	0.0000	0.0648	0.0918	0.0270
Corse	3	0.1890	0.0000	0.0926	0.0930	0.0004
Limousin	3	0.4650	0.4255	0.2590	0.1051	-0.1539
Alsace	2	0.2477	0.0000	0.0859	0.0936	0.0077
Champagne-Ardenne	2	0.2518	1.0989	0.3622	0.1055	-0.2566
Franche-Comte	1	0.1962	0.0000	0.0918	0.0950	0.0032

At $N = 500$ the comparison is dominated by sampling artifacts rather than genuine model disagreement. Twelve of 21 regions record zero claims in the subsample, producing raw rates of zero and Bühlmann estimates that are essentially $(1 - Z_r) \times \bar{Y}$ - pure shrinkage toward the global mean with no regional signal. At the other extreme, Champagne-Ardenne draws one or two claims on two policies, inflating its raw rate to 1.099 and its Bühlmann estimate to 0.362; even with a credibility factor of only 0.252, the extreme raw rate is enough to pull the estimate far above any plausible regional risk. The hierarchical model is insulated from both pathologies by the covariate structure and the prior on σ_u , which prevent any single region from straying far from a plausible range. The large differences in the rightmost column at $N = 500$ therefore measure not the distance between the two frameworks but the distance between data reliability and model-stabilised inference. This distinction collapses as N grows and the zero-claim regions accumulate enough exposure to produce stable rates.

Table 8: Bühlmann vs. Hierarchical Posterior Means at $N = 2000$

Region	n	Z_r	Raw Rate	Bühlmann	Hierarchical	Difference
Centre	474	0.9478	0.0850	0.0854	0.0947	0.0093
Rhone-Alpes	250	0.8839	0.0780	0.0798	0.0891	0.0093
Provence-Alpes-Cotes-D’Azur	234	0.8689	0.1344	0.1290	0.1003	-0.0287
Ile-de-France	206	0.8385	0.1143	0.1110	0.0914	-0.0196
Bretagne	124	0.8358	0.0933	0.0933	0.0932	-0.0001
Nord-Pas-de-Calais	119	0.7685	0.0715	0.0766	0.0891	0.0125
Pays-de-la-Loire	114	0.8060	0.0857	0.0872	0.0927	0.0055
Languedoc-Roussillon	105	0.7005	0.0254	0.0458	0.0871	0.0413
Aquitaine	92	0.6955	0.1299	0.1189	0.0964	-0.0225
Poitou-Charentes	56	0.6394	0.1004	0.0980	0.0938	-0.0042
Midi-Pyrenees	50	0.5727	0.1328	0.1161	0.0953	-0.0208
Basse-Normandie	32	0.5574	0.0943	0.0940	0.0926	-0.0014
Bourgogne	31	0.4739	0.0659	0.0805	0.0920	0.0115
Haute-Normandie	26	0.3579	0.1065	0.0982	0.0937	-0.0045
Picardie	24	0.3601	0.0000	0.0599	0.0897	0.0298
Auvergne	16	0.2469	0.0000	0.0705	0.0921	0.0216
Corse	13	0.2170	0.2141	0.1198	0.0948	-0.0250
Limousin	13	0.3139	0.1297	0.1049	0.0942	-0.0107
Champagne-Ardenne	9	0.1638	0.3030	0.1279	0.0955	-0.0324
Alsace	6	0.1765	0.0000	0.0771	0.0923	0.0152
Franche-Comte	4	0.1224	0.4255	0.1342	0.0957	-0.0385

At $N = 2,000$ the comparison stabilises considerably. The zero-claim pathology is now confined to three regions (Picardie, Auvergne, Alsace) rather than twelve, and the extreme raw rates that destabilised Champagne-Ardenne and Limousin at $N = 500$ have balanced, though Champagne-Ardenne’s raw rate of 0.303 and Franche-Comté’s 0.426 remain unreliable on their small samples. Credibility factors of 0.164 and 0.122 correctly assign them little weight, pulling their Bühlmann estimates back toward the global mean. Where the data is adequate — the eight largest regions each with over 100 policies. The Bühlmann and hierarchical estimates are already close, with differences typically below 0.015. The most persistent source of issue is Languedoc-Roussillon, whose raw rate of 0.025 is implausibly low given its full-sample rate of 0.105; the Bühlmann estimator partially corrects this to 0.046 while the hierarchical model, returns 0.087. This pattern, the hierarchical model outperforming Bühlmann in regions whose subsample composition is unrepresentative of their true risk profile, is the consequence of conditioning on covariates, and it is exactly what the approximation caveat anticipates.

Table 9: Bühlmann vs. Hierarchical Posterior Means at $N = 10000$

Region	n	Z_r	Raw Rate	Bühlmann	Hierarchical	Difference
Centre	2372	0.9509	0.1033	0.1034	0.1153	0.0118
Rhone-Alpes	1250	0.8960	0.1072	0.1071	0.0992	-0.0079
Provence-Alpes-Cotes-D’Azur	1168	0.8705	0.1411	0.1365	0.1120	-0.0245
Ile-de-France	1030	0.8506	0.1328	0.1288	0.0939	-0.0349
Bretagne	622	0.8433	0.0858	0.0889	0.0977	0.0088
Nord-Pas-de-Calais	594	0.7884	0.0757	0.0821	0.0904	0.0083
Pays-de-la-Loire	572	0.8061	0.1172	0.1150	0.1067	-0.0084
Languedoc-Roussillon	526	0.7450	0.1009	0.1022	0.0992	-0.0030
Aquitaine	462	0.7175	0.0909	0.0952	0.0958	0.0007
Poitou-Charentes	281	0.6837	0.0771	0.0863	0.0962	0.0099
Midi-Pyrenees	252	0.5660	0.1573	0.1351	0.1065	-0.0286

Region	n	Z_r	Raw Rate	Bühlmann	Hierarchical	Difference
Basse-Normandie	161	0.5736	0.0667	0.0835	0.0945	0.0111
Bourgogne	155	0.4894	0.0803	0.0934	0.0969	0.0035
Haute-Normandie	130	0.3605	0.0227	0.0760	0.0926	0.0166
Picardie	118	0.4205	0.1413	0.1209	0.1033	-0.0176
Auvergne	78	0.2817	0.0654	0.0946	0.0977	0.0031
Corse	67	0.2429	0.1998	0.1288	0.1054	-0.0234
Limousin	67	0.2748	0.0338	0.0862	0.0961	0.0099
Champagne-Ardenne	45	0.1908	0.1631	0.1170	0.1025	-0.0144
Alsace	32	0.1633	0.0000	0.0887	0.0951	0.0063
Franche-Comte	20	0.0970	0.1193	0.1074	0.1002	-0.0072

At $N = 10,000$ the sampling problems that dominated the smaller subsamples have largely resolved. Only Alsace records zero claims, and the raw rates across the remaining 20 regions are broadly consistent with their full-sample values. The credibility factors now span a meaningful range, from 0.097 for Franche-Comté (20 policies) to 0.951 for Centre (2,372 policies), and the Bühlmann estimator is doing genuine work rather than simply collapsing toward the global mean. Where Z_r is high, the eight largest regions, the two estimators agree closely, with differences typically below 0.012. The remaining disagreements are concentrated in two places: regions with genuinely extreme underlying rates (Provence-Alpes-Côtes-d’Azur at 0.112 hierarchical vs. 0.139 Bühlmann, Midi-Pyrénées at 0.107 vs. 0.145) where the covariate adjustment pulls the hierarchical estimate back toward the global mean, and regions small enough like Haute-Normandie and Limousin that the subsample remains unrepresentative. The overall mean absolute difference has fallen substantially relative to $N = 2,000$, and the two frameworks are now telling the same story about which regions are genuinely high and low-risk, differing only in the magnitude of the deviation they attribute to each.

Table 10: Bühlmann vs. Hierarchical Posterior Means at $N = 676k$

Region	n	Z_r	Raw Rate	Bühlmann	Hierarchical	Difference
Centre	160529	0.9959	0.0897	0.0897	0.1046	0.0149
Rhone-Alpes	84592	0.9908	0.1112	0.1111	0.1074	-0.0037
Provence-Alpes-Cotes-D’Azur	79019	0.9883	0.1099	0.1098	0.1023	-0.0075
Ile-de-France	69680	0.9862	0.1319	0.1315	0.1012	-0.0303
Bretagne	42109	0.9851	0.0974	0.0975	0.1058	0.0083
Nord-Pas-de-Calais	40168	0.9789	0.0915	0.0917	0.0881	-0.0036
Pays-de-la-Loire	38729	0.9812	0.0917	0.0919	0.0986	0.0066
Languedoc-Roussillon	35597	0.9718	0.1058	0.1056	0.1013	-0.0043
Aquitaine	31274	0.9714	0.0944	0.0946	0.0940	-0.0006
Poitou-Charentes	19035	0.9637	0.0868	0.0873	0.0958	0.0085
Midi-Pyrenees	17061	0.9428	0.0985	0.0986	0.0938	-0.0049
Basse-Normandie	10873	0.9404	0.0953	0.0956	0.0998	0.0042
Bourgogne	10470	0.9225	0.1018	0.1017	0.1001	-0.0016
Haute-Normandie	8772	0.8828	0.0954	0.0961	0.0967	0.0006
Picardie	7985	0.8945	0.1232	0.1208	0.1072	-0.0137
Auvergne	5279	0.8463	0.0817	0.0846	0.0872	0.0025
Limousin	4566	0.8507	0.1082	0.1071	0.1141	0.0070
Corse	4516	0.8077	0.1399	0.1324	0.1117	-0.0207
Champagne-Ardenne	3020	0.7403	0.1335	0.1251	0.1107	-0.0143
Alsace	2192	0.7407	0.1108	0.1082	0.1001	-0.0081
Franche-Comte	1323	0.5716	0.1016	0.1013	0.0988	-0.0026

At $N = 676,789$ the comparison reaches its clearest form. Credibility factors are uniformly high, ranging

from 0.572 for Franche-Comté to 0.996 for Centre, meaning every region is being trusted heavily over the global mean, and the Bühlmann estimates are essentially identical to the raw rates. The structural zeros, extreme subsamples, and compositional imbalances that complicated the smaller samples have vanished entirely. What remains is a clean, interpretable result: the systematic gap between the Bühlmann estimate and the hierarchical posterior mean is now driven almost entirely by the covariate adjustment. Île-de-France is the clearest example, with a raw rate of 0.132 and Bühlmann estimate of 0.132 both reflect a genuinely high-risk region, but the hierarchical model returns 0.101, suggesting that much of Île-de-France’s excess claims are explained by its portfolio composition (younger drivers, higher bonus-malus scores, higher population density) rather than regional risk beyond what the covariates capture. The same logic applies in reverse to Centre, whose hierarchical estimate of 0.105 exceeds its raw rate of 0.090, implying its portfolio skews toward lower-risk profiles. At full sample the Bühlmann-hierarchical difference is therefore not noise but signal; it measures precisely the contribution of covariate adjustment to regional rate estimation, and it is the most honest version of the approximation caveat stated earlier.

7 Summary and Conclusions

This analysis set out to answer whether a hierarchical Bayesian modeling improves estimation of regional claim frequencies relative to a flat pooled model, and whether the choice of prior on regional variance matters? For the French motor insurance portfolio the empirical answer is clear, but the more interesting result is what the analysis reveals about hierarchical modeling as a general inferential framework.

The direct answer to the research question is yes; partial pooling delivers better regional estimates than complete pooling at every sample size examined, and it does so in a way that is interpretable through the classical Bühlmann credibility framework. At full sample the posterior mean of σ_u converges to approximately 0.085 under both priors, implying that genuine regional risk varies by roughly $e^{0.085} \approx 9\%$ around the global mean once covariates are accounted for. This residual regional heterogeneity is real and structurally meaningful, it is not accounted for by driver age, vehicle characteristics, or population density, and the hierarchical model captures it while the flat model remains blind to it.

We also see the interplay between sample size, prior choice, and the identifiability of σ_u . At $N = 500$ the informative prior dominates inference entirely, effectively imposing complete pooling regardless of model specification, as the likelihood is too weak to overcome the tight prior belief that regions are homogeneous. By $N = 2,000$ the data begins to push back, by $N = 10,000$ the two hierarchical posteriors are visibly converging, and at full sample size the distinction between the two priors is negligible. The σ_u posterior densities across sample sizes suggest that prior sensitivity diminishes rapidly with increasing group-level information, though the precise threshold of this transition was not systematically identified in this analysis.

The deeper reason for this behaviour is a structural feature of hierarchical models that generalises far beyond this dataset. σ_u is identified by the number of groups, not the number of observations. In this analysis that number is 21, and it does not grow as N increases. Even at full sample size, the prior on σ_u is being updated by essentially 21 sufficient statistics (very precise ones, but 21 nonetheless). This group-level bottleneck applies wherever hierarchical models are used: meta-analyses with few studies, longitudinal designs with few subjects, multilevel models with sparse higher-level units, etc. This implies that prior choice on variance components matters most precisely when it is least verifiable, in the small-portfolio setting where the data is not sufficiently large to distinguish between competing beliefs about between-group heterogeneity. This is the heart of Bayesian analysis, using prior beliefs to supplement a model when the data is too sparse to speak for itself.

The Bühlmann comparison sharpens this conclusion, separating the sources of shrinkage that the hierarchical model combines. At small N , differences between the Bühlmann and hierarchical estimates are dominated by sampling properties such as zero-claim regions, and extreme raw rates on one or two policies. At full sample, these vanish and a clean residual emerges: the systematic gap between the two estimators measures precisely the contribution of covariate adjustment to regional rate estimation. Île-de-France’s apparent excess risk shrinks once its portfolio composition is conditioned on; Centre’s apparent deficit reverses for the same reason. Classical Bühlmann, operating on raw group means, cannot see this, and it attributes the entire

deviation from the global mean to genuine regional risk. The hierarchical model differentiates that deviation into a covariate-explained component and a regional effect, which is more informative.

We can make general conclusions beyond this analysis. Hierarchical Bayes and credibility theory are not competing frameworks but complementary ones, with the former incorporating the latter while adding covariate adjustment, quantifying posterior uncertainty, and inferring on prior beliefs about between-group variance. Additionally, prior sensitivity analysis is most valuable precisely where it is most uncomfortable, not as a robustness check, but as an acknowledgment that when data is insufficiently informative, the analyst's subjective beliefs about between-group heterogeneity are the dominant driver of inference. In such settings the choice of prior on σ_u is not a technical detail but a substantive actuarial judgment about how similar we expect regions to be before analysing the data.